

- Solar Panels

- They absorb sunlight and convert it into usable electricity, which will be used to help power the building they're a part of, and in some setups unused energy is stored in a battery to be used in emergencies or when there is no sunlight.
- I would recommend them in all homes, buildings, and cities as it generates effectively free electricity, while it might not be able to power all electronics, it would still significantly lower electricity bills, as you would be less dependent on the grid. It also is clean, it creates no emissions and also reduces your carbon footprint, as some energy providers may use polluting energy sources under high demand, which using a solar panel reduces the demand on the grid.
- As like many things, they can fail, they require proper maintenance and cleaning, without this they can start to fail, either completely or with a reduced output. They also work best in areas where there is direct sunlight, which could be easily blocked by a new building, or they may not output very much if you live in an area that naturally does not get a lot of sun. You could overcome some of these issues by keeping on top of maintenance, and making sure the panels are clean and clear of any obstructions, and before installation contact the local council to see if there could be a possibility of something being built in the future that could obstruct them.

- LED Lights

- They work like regular lights, however they are much more efficient than incandescent, as they do not use heat to create light, they are directly converted to light using semiconductors, which also takes up much less energy for the same amount of brightness, which lowers the costs to

run them. They also last much longer than incandescent as they are less susceptible to the failures that they had.

- I recommend them in all environments because it lowers costs and emissions, this is because they require less energy, lowering electric costs, they last longer reducing maintenance, and reduce emissions because the lower demand on the grid means there's no need to run polluting electricity sources.
- You could face issues when swapping an old incandescent light with an LED light in some scenarios, old dimmer systems are unlikely to work with LED lights and can cause flickering when being dimmed. You could workaround this by checking if your light or dimmer work with each other. Another issue is LED lights can be quite "harsh" as in they're too cold (white) which can make people feel uncomfortable, the workaround would be to get LED lights that have a warm white colour like incandescent lights.
- Smart thermostats
 - They're very similar to regular thermostats, except they learn usage patterns, and analyse other data to match with it, which it uses to automatically adjust the temperature for you for when it thinks you will need it and when you won't, which helps cut down on wasted resources, as it handles all of the users guess work on what to set it for and how long.
 - I'd recommend having them in all buildings because it would help reduce costs and emissions as they adapt the heating with all factors included, it would reduce your carbon footprint as it can turn off heating when it is not needed, unlike being done manually or through older thermostats.

- You could face a problem if you have an internet outage, losing access remotely, and you could lose some advanced functionality on the thermostat as some of its data may rely on external resources. You could only overcome this through having a backup internet connection.

AI and robotics do have an impact on the job market and in society, they can replace some peoples existing jobs, like for example, a worker that is designated to visually observe objects on a conveyer to see if there is any defects, and if there is one it must be removed. Robotics/AI could easily replace this job as it can be much more “efficient” and a person could very easily get fed up of doing this task all day, and could quit, while an AI/Robot cannot do that. It affects society because it can remove many job opportunities as companies might find tech can do the job better than a person, leaving more and more people out of work, struggling to find somewhere new and making rent.

This tech does benefit companies as it means there are less people to pay, and AI can open doors as it gives knowledge and useful data to those who don't even understand what they're asking it, though that can be risky. Though linking back, many people that seek jobs are left out, as more and more job roles are cut as they were found that AI/Robotics could do it better for less.

Advantages

- Autonomous
- Large amounts of knowledge
- Doesn't need breaks

Disadvantages

- Power hungry
- AI can have privacy concerns
- Might be worse than an actual person

Making a computer more efficient could be done through changing some of its components, you could replace the CPU for a more modern, or efficiency targeted model, you could replace the PSU for a more modern one, if it is not already, and if you don't need it you could remove your GPU and use the integrated graphics, if needed you could swap it for a low power model.